

Andrei Silviu Dragnea

Education

2013-2017 **BSc, Computer Science and Engineering**; Faculty of Automatic Control and Computers, Politehnica University of Bucharest (Bucharest)
*Thesis title: Automatic Recursion Removal in Java (as an **IntelliJ Plugin**)*

Experience

- Expert in refactoring and decoupling complex systems into self-contained and loosely-coupled components
- IntelliJ plugin **developer**
- **Spring** and **Spring Boot** framework expert
- Async programming expert: **Project Reactor** from Spring, **Kotlin coroutines**, **cats-effect** and **zio** from Scala, Rust **async** programming
- Experience in profiling JVM applications and optimizing runtime performance and resource usage using **VisualVM**
- Open source contributor to **Docker**, **mockito**, **intellij-kotlin**, **intellij-community**, **reactor-core**, **reactor-netty**, **netty**, **spring-framework**, Scala **kubernetes-client**, **Renovate**
- Proficient in Java, Kotlin, Scala, Rust, C and Python
- Build system expert: Maven, Gradle, Earthly, Docker, sbt. Reduced build times of Scala projects significantly.
- Author of an automatic Git **rebaser** written in Rust
- Passionate about static analysis and automatic refactoring in Java, Kotlin, Scala and Rust
- Working on a **converter** from Java reactive code to Kotlin coroutines
- Experience with enterprise identity protocols and frameworks, such as OAuth2, SCIM2, SAML
- Worked with Kafka, Docker, Kubernetes, Envoy, Grafana, Prometheus, AWS EC2, S3, Route53, Cloudwatch, Splunk, Datadog, NewRelic, Azure Active Directory, Jenkins
- Good knowledge of Operating Systems internals and how async frameworks are implemented
- Promoter of the **clean tests** paradigm

May 2021 - Present **Software Engineer at Adobe (Media Analytics team)** I work on a performance-sensitive system collecting Analytics events from video players in web browsers. I optimized the runtime performance by correctly handling blocking calls, thus avoiding starving the **cats-effect** compute pool. I also enforced strong static analysis checks on the Scala project at compile time, in order to maintain the high quality of the code. I re-wrote the event ingestion system in Rust as a POC, resulting into 2.6x CPU usage and 50x memory usage reduction. I also optimized Kafka Java client performance by using a separate thread for blocking `KafkaProducer.send()` calls.

March 2018 - May 2021 **Software Engineer at Adobe (Identity Management Services team)** I was part of the Identity Management Services team. I worked on implementing the server

side of the SCIM2 protocol over a Spring 5 reactive stack. I developed a thin wrapper over Mockito and Spring TestContext framework in order to promote writing **clean tests**. I introduced Kotlin and coroutines to our team, as a developer-friendly alternative to writing non-blocking code. I made major software architectural changes by using smart refactoring tricks, resulting in a simpler system design.

**September
2017 - De-
cember 2017**

Software Engineer at Instacar I implemented a TCP server over a proprietary binary protocol for remote car control. I recompiled the Linux kernel with custom TCP settings in order to compensate for a faulty TCP implementation on the devices used for remote car control. I introduced type hints to the Python 3 codebase and I developed a Pycharm **plugin** for dead code elimination.

I always strive to find the simplest solution possible to a problem. I proactively fix problems from any software project that I encounter, whether it is internal to my company, or an open source project.

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